

Shashank Singh

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EDUCATION

University of Lucknow

B.Tech in Computer Science and Engineering (AI)

Lucknow, UP

Oct. 2023 – Present

Einstein Public School

Intermediate (Class XII) | 86.8%

Pratapgarh, UP

Apr. 2021 – July 2022

Einstein Public School

High School (Class X) | 90%

Pratapgarh, UP

Apr. 2019 – July 2020

EXPERIENCE

Data Science Intern

June 2025 - July 2025

NIELIT (Certificate)

Gorakhpur, UP

- Performed data preprocessing, feature engineering, and EDA on 5+ real-world datasets using NumPy, Pandas, and Matplotlib; handled missing values, categorical encoding, and outlier removal across 6+ features per dataset.
- Implemented and evaluated 6 ML algorithms – Linear Regression, Logistic Regression, Decision Trees, SVM, Random Forest, and KNN – using Scikit-learn; assessed models using R-squared, F1 score, precision, and recall.
- Built 10+ interactive visualizations using Matplotlib (bar, scatter, histogram, heatmap) to identify trends and correlations, directly informing feature selection and model decisions.

PROJECTS

VectorDB | C++, HNSW, KD-Tree, RAG, Ollama, cpp-httplib, HTML/CSS/JS

- Architected a vector database from scratch in C++, implementing HNSW, KD-Tree, and Brute-Force search side-by-side across 3 distance metrics; benchmarked all three via a dedicated REST endpoint, contrasting $O(\log N)$ graph search against $O(N)$ brute force.
- Engineered a full local RAG pipeline via Ollama – chunking documents into overlapping 250-word segments, embedding them (768D nomic-embed-text), and retrieving context for a local Large Language Model (Llama 3.2) – exposed through a full REST API with a live 2D PCA visualization frontend.

Conversational RAG Chatbot | Streamlit, LangChain, ChromaDB

Live Demo

- Developed a RAG web app using Streamlit, LangChain, and Groq that lets users upload PDFs and hold multi-turn conversations grounded in their content; a history-aware retriever reformulates follow-up questions using prior chat context.
- Chunked and embedded documents with HuggingFace's all-MiniLM-L6-v2, indexed them in ChromaDB for semantic search, and served fast inference via Groq's Llama 3.3 70B model, with support for multiple isolated chat sessions.

Sports Person Classifier | Python, OpenCV, Scikit-learn, Flask, Wavelet Transform

Live Demo

- Designed a full-stack Image Classification pipeline using OpenCV Haar Cascades for face detection and Wavelet Transform for feature extraction; achieved 87%+ accuracy by tuning SVM, Random Forest, and Logistic Regression models via GridSearchCV.
- Exposed predictions through a RESTful Flask API and designed an interactive frontend supporting real-time image upload with sub-second inference response time.

TECHNICAL SKILLS

Languages: Python, C++, SQL, HTML, CSS, JavaScript

AI/ML: Scikit-learn, OpenCV, RAG, Vector Databases(HNSW), Ollama, Feature Engineering, EDA, Model Deployment

Frameworks & Tools: Flask, Streamlit, Pandas, NumPy, Git, GitHub, VS Code, Jupyter

CERTIFICATIONS

AWS Training & Certification - Fundamentals of ML & AI

May 2026 – June 2026

Samsung Innovation Campus - Artificial Intelligence

Oct. 2025 – Dec. 2025